MILITARY SPECIFICATION

ANTENNA AS-1304( )/ARN-30

1. SCOPE

1.1 This document covers one type of Very High Frequency (VHF) Omni-directional range balanced loop airborne antenna for the reception of Instrument Landing System (ILS), Localizer and Very High Frequency Omni-range (VOR) signals for use with Airborne Navigational equipments within the frequency range of 108 to 122 MHz and designated AS-1304( )/ARN-30 (See 6.1 and 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet-Materials Handling Wood (General Construction Requirements)
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed & Lock Corner
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, Corrugated & Solid Sheet Stock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-45	Tape, Gummed Paper, Reinforced and Plain, for Sealing & Securing

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PPP-T-76	Tape, Pressure Sensitive Adhesive Paper, Water Resistant (For Carton Sealing)
PPP-T-97	Tape, Pressure Sensitive, Filament Reinforced
QQ-S-781	Strapping, Steel, Flat and Seals

MILITARY

MIL-P-116	Preservation, Method of
MIL-E-5400	Electronic Equipments, Aircraft, General Specification for
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

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DRAWINGS

USAECOM

DL-SM-A-498900 Antenna Assembly AS-1304( )/ARN-30

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. - Antenna AS-1304( )/ARN-30 is a horizontally polarized, lightweight VHF omnidirectional, balanced loop, airborne antenna. It is used with equipment receiving ILS, Localizer and VOR signals within the VHF range of 108 to 122 MHz and designed for light planes and helicopters.

3.2 First Article Samples. - The contractor shall furnish three pre-production samples for approval, if required by the invitation for bids and contract. (See 4.3 and 6.2)

3.3 Construction. - The units on order shall be constructed in accordance with the following drawing and data list and its associated drawings: DL-SM-A-498900.

3.4 Cleaning.

3.4.1 Parts. - After fabrication, parts shall be cleaned in accordance with Requirement 9 of MIL-STD-454, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.4.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

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3.5 Finish, Protective. The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. The final paint film on Type 1 surfaces shall be FILM B, in accordance with MIL-F-14072. (See 4.4)

3.6 Securing of Parts. -

3.6.1 General. - Brackets, lugs, inserts, bolts and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.7 Marking. -

3.7.1 General. - Marking shall conform to MIL-M-13231 and the Equipment Drawings. (See 4.4)

3.7.2 Visibility. - Whenever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.7.3 Serial Numbers. - The unit(s) on this order shall have serial numbers.

3.8 Soldering. -

3.8.1 Solder. - Solder shall conform to Requirement 5 of MIL-STD-454 and the equipment drawings.

3.8.2 Acid or Acid Salts. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.8.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means

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employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the unit. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.8.4 Clearance. - Clearance between solder connections or bare conductors, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.9 Connections. -

3.9.1 General. - Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.10 Tropicalization of Materials. - Materials shall be treated as follows:

3.10.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.10.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.10.3 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

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3.10.4 Statement of Treatment. - The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use.

3.11 Service Conditions (See 4.7). - The unit(s) shall meet the service conditions specified in MIL-E-5400 for Class 2 equipment, as follows:

3.11.1 Temperature-Altitude (See 4.7.1). - Antenna performance shall be as indicated for ambient temperatures from -54°C to +95°C at altitudes from sea level to 70,000 feet. The antenna shall withstand extended exposure to temperatures within the -62°C to +71°C range.

3.11.2 Humidity (See 4.7.2). - The antenna shall withstand relative humidity conditions in excess of 95% and shall be capable of satisfactory performance, within 1/2 hour after subjection to the test.

3.11.3 Vibration (See 4.7.3). - The antenna shall perform satisfactorily after subjection to vibration environments in accordance with the test of 4.7.3.

3.11.4 Shock (See 4.7.4). - The antenna shall perform satisfactorily after subjection to a shock environment in accordance with the shock test of 4.7.4.

3.11.5 Acceleration (See 4.7.5). - The unit(s) shall withstand 6g steady state acceleration for one minute.

3.11.6 Rain (See 4.7.6). - The unit(s) shall meet the performance requirements of this document when subjected to rain as encountered in service, with the base down, in accordance with Method 506 of MIL-STD-810.

3.11.7 Salt Fog (See 4.7.7). - The unit(s) shall be exposed, in normal operating configuration, to salt fog for 48 hours and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish.

3.11.8 Sand and Dust (See 4.7.8). - The unit(s) shall withstand exposure to sand and dust particles as encountered in desert areas. It shall show no evidence of corrosion as evidenced by any visible degradation of material and finish.

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3.12 Electrical Requirements (See 4.8). -

3.12.1 Output Impedance (See 4.8.1). - The unit(s) feed ports shall have a 50 ohm output impedance from 108 to 122 MHz.

3.12.2 Voltage Standing Wave Ratio (VSWR) (See 4.8.2). - The impedance of the antenna at the input conector of the cable assembly, shall be such that the VSWR referenced to a 50-ohm system does not exceed 5.0 to 1 throughout the frequency range of the antenna. The measurement shall be made with the antenna mounted on the ground plane of Figure 3.

3.12.3 Polarization (See 4.8.3). - The antenna shall receive horizontally polarized signals. Reception of vertically polarized signals shall be at least 12 dB down from the level of horizontally polarized signals. The measurement shall be made with the antenna mounted on the ground plane of Figure 3.

3.12.4 Radiation Pattern (See 4.8.3). - The radiation field pattern shall be essentially omnidirectional in the azimuth plane and shall approximate a Cos θ curve in the vertical plane.

3.12.5 Gain (See 4.8.4). - Antenna gain relative to an isotropic source shall not be less than -2 dB.

3.13 Operational Inspection. - Each unit(s) shall be capable of meeting the specified test after applicable processes, inspection and test have been completed to determine that each unit(s) on order is operable prior to packaging. (See 4.11)

3.14 Interchangeability (See 4.10). - Like units, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Components of the antenna shall not be designed as matched units. Antenna half loops shall be completely interchangeable and shall perform as indicated in 3.13 with any cable assembly in Drawing Number DL-SM-A-498900.

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3.15 Workmanship. - The unit(s) shall be manufactured and assembled in accordance with requirement nine (9) of MIL-STD-454 and the applicable portions of the following paragraphs herein:

Cleaning	3.4
Securing of parts	3.6
Soldering	3.8
Connections	3.9
Tropicalization of materials	3.10

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. - Inspection shall be classified as follows:

a. First Article inspection (does not include preparation for delivery (See 4.3)).

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.12 and 4.13)

4.3 First Article Inspection. - Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B and Group C (See Tables I, II and III respectively). The preproduction inspection will normally be performed in this order: (1) vibration, (2) shock. Other first article inspection may precede, follow, or be interspersed between the foregoing.

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4.4 Inspection Covered by Subsidiary Documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order:

ITEM	WHERE REQUIRED
Finish protective	3.5
Marking	3.7

4.5 Quality Conformance Inspection of Equipment Before Preparation for Delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performance of any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.7) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A Inspection. - Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table I.

TABLE I - GROUP A INSPECTION

Inspection	Reqt Par	Insp Par	AQL Major	AQL Minor
Visual and Mechanical. -	3.15	4.9		
AS-1304( )/ARN-30:				
Base Assemblies (less top)			0.65%	2.5%
Base Assemblies (assembled)			1.0%	4.0%
AS-1304( )/ARN-30: (assembled)			1.0%	4.0%
Electrical:				
Voltage Standing Wave Ratio	3.12.2	4.8.2	*1% for the combined Group	
Polarization	3.12.3	4.8.3		
Operational Inspection	3.13	4.11		0.1%

*All electrical defects shall be considered major defects.

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4.5.2 Group B Inspection. - Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table II and to the special inspection levels of Table I of MIL-STD-105.

Group B sampling plans. - The Group B AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal tightened and reduced inspection.

TABLE II - GROUP B INSPECTION

Inspection	Reqt Para	Insp Para	AQL
Output Impedance	3.12.1	4.8.1	6.5%
Interchangeability	3.14	4.10	6.5%

4.5.3 Group C Inspection. - Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table III. Samples shall be selected in accordance with 4.5.3.1.

TABLE III - GROUP C INSPECTION

Inspection	Reqt Para	Insp Para
Subgroup I:		
Temperature/Altitude	3.11.1	4.7.1
Humidity	3.11.2	4.7.2
Vibration	3.11.3	4.7.3
Radiation Pattern	3.12.4	4.8.3
Gain, Antenna	3.12.5	4.8.4
Subgroup II:		
Shock	3.11.4	4.7.4
Acceleration	3.11.5	4.7.5
Rain	3.11.6	4.7.6
Salt Fog	3.11.7	4.7.7
Sand and Dust	3.11.8	4.7.8

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4.5.3.1 Sampling for Group C inspection. - Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

4.5.3.1.1 Subgroup I: - For this subgroup, two units from the first 25 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two units each month shall be selected.

4.5.3.1.2 Subgroup II. - For this subgroup, two units from the first 25 production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units every two months shall be selected.

4.5.3.2 Order of inspection within Group C. - Group C testing shall be performed in accordance with Table IV in the sequence shown. The items shall be cleaned, dried and refurbished if required before proceeding to the next test.

TABLE IV - GROUP C INSPECTION

INSPECTION SUBGROUP I	INSPECTION SUBGROUP II
Temperature/Altitude	Rain
Radiation Pattern	Sand and Dust
Gain, Antenna	Salt Fog
Humidity	Acceleration
Vibration	Shock

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4.5.4 Reinspection of Conforming Group C Sample Units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

4.6 Standard Test Conditions. - Unless otherwise specified, tests shall be performed under prevailing ambient conditions of temperature, pressure and relative humidity. The following test conditions shall apply when specified herein:

a. Cylindrical ground plane (See Figure 3).

(1) Material: Aluminum

(2) Thickness: 1/8 inch.

(3) Width: Approximately 21 inches

(4) Length: Approximately 132 inches

b. Load: 50 ohms

c. Test meter bridge, Hewlett Packard HP803AVHF or equal.

4.7 Service Conditions. - The unit(s) shall be subjected to and meet the service conditions in accordance with MIL-STD-810 as Class 2 equipment, except as defined herein. The electrical test of 4.8.2 shall be conducted prior to and subsequent to each service condition that requires a test to be conducted. The measurement in this test shall not change more than five (5%) percent from the value measured prior to subjection to the test condition; in no event shall the measurement exceed the specified limit. VSWR shall be measured during the temperature and humidity tests while antenna is in chamber, in addition to tests prior and subsequent to these environmental tests. Any type of chamber of adequate size to hold the specified ground plane will suffice for these tests since only a possible change in the VSWR is looked for. A pre-test VSWR is to be made to obtain a chamber reference VSWR.

4.7.1 Temperature-altitude test. - The temperature altitude test of MIL-STD-810, Method 504, shall be performed, except that the antenna shall be non-operating during application of environment. A VSWR performance test, in accordance with 4.8.2 shall be performed at the end of step 4 and at the completion of the test. (See 3.11.1)

4.7.2 Humidity test. - The humidity test of MIL-STD-810, Method 507, Procedure II, shall be performed. A VSWR test in accordance with 4.8.2 shall be performed as soon as practicable after the completion of the test, but not later than 1/2 hour after completion. (See 3.11.2).

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4.7.3 Vibration test (See 3.11.3). - The vibration test of MIL-STD-810, Method 514.1, Procedure I curve Z, part 1 and 2, shall be performed. At the completion of the test, the VSWR performance test of 4.8.2 shall be performed.

4.7.4 Shock test. - The equipment shock test of MIL-STD-810, Method 516.1, Procedure I, shall be performed. At the completion of the test, the VSWR performance test of 4.8.2 shall be performed. (See 3.11.4)

4.7.5 Acceleration test. - The unit(s) shall be tested in accordance with Method 513.1, Procedure II of MIL-STD-810, with a test level of 6g and a test time of 1 minute.

4.7.6 Rain test. - The unit(s) shall be tested in accordance with Method 506, MIL-STD-810. The antenna shall be mounted with base down for demonstrating compliance with 3.11.6. At the completion of the test, the VSWR performance test of 4.8.2 shall be performed.

4.7.7 Salt fog test. - The Salt Fog Test of MIL-STD-810, Method 509, shall be performed. The antenna connector may be plugged to simulate installed condition. The unit shall not be required to operate electrically. The units shall meet the requirement of 3.11.7.

4.7.8 Sand and Dust test. - The unit(s) shall be subjected to the sand and dust test of Method 510, MIL-STD-810, for compliance with 3.11.8. At the completion of the test, the VSWR performance test of 4.8.2 shall be performed.

4.8 Electrical.

4.8.1 Output Impedance. - Set the Hewlett-Packard (H.P.) 803AVHF bridge or equal test meter to the required frequency and make required adjustments. Connect the test meter to the feed port. Measurements shall be made to determine compliance with 3.12.1.

4.8.2 Voltage Standing Wave Ratio (VSWR) test. - VSWR measurements shall be made under standard conditions. The VSWR shall be measured with the antenna installed on the ground plane of Figure 3. Measurements shall be made in increments of 2 MHz throughout the frequency range. After setting the meter to the required frequency, calculate the VSWR on a "Z-θ" chart for compliance with 3.12.2. This measurement technique shall have prior approval by the procuring activity.

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4.8.3 Pattern measurements. - Radiation pattern measurements shall be made for both horizontal and vertical polarization of the source antenna. Measurements shall be made with the antenna mounted on the ground plane of Figure 3 at increments of 2 MHz throughout the frequency range: (See 3.12.4) This measurement technique shall have prior approval by the procuring activity.

4.8.4 Gain, antenna. - The gain of the antenna shall be measured relative to an isotropic source. The antenna shall be tested to determine compliance with 3.12.5. The measurement technique shall have prior approval by the procuring activity.

4.9 Visual and Mechanical. - The assemblies, prior to assembly, and the complete assembled unit(s) shall be examined for defects in accordance with MIL-STD-252.

4.10 Interchangeability. - The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.14. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of moldings, mounting bases, and insertable parts, when such dimensions affect mating.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Locations of holes on separable parts which must mate such as mounting assemblies.

d. Location of connectors, mounting base holes and elements.

e. Size and form of threads.

4.11 Operational Inspection. - Each unit shall be inspected for assurance that, when fed a modulated RF signal between 108 and 122 MHz and the demodulated signal is displayed on an oscilloscope, there shall be no interruptions or noise in the displayed signal as the unit(s) is vibrated at 5g's between 25 and 55 Hz. (See 3.13)

4.12 Quality Conformance Inspection of Preparation for Delivery. - Preparation for delivery shall be inspected in accordance with MIL-P-116, to determine conformance to the packaging requirements.

4.13 Rough Handling Test (Preparation for Delivery). - When rough handling test in accordance with MIL-P-116 is required by the contract, the operational inspection test 4.11 shall be conducted to determine freedom from operational malfunction caused by rough handling.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be Level A or C, as specified (See 6.2 and 6.5).

5.1.1 Level A. -

5.1.1.1 Cleaning. - Antenna AS-1304( )/ARN-30, shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Antenna AS-1304( )/ARN-30 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116, as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged Method IC-1.

5.1.1.4.2 Antenna AS-1304( )/ARN-30. - Each Antenna AS-1304( )/ARN-30, consisting of two half loop antennas and one Cable Harness assembly, shall be packaged Method III, as follows: Cushion each half loop antenna by wrapping in paperboard conforming to PPP-P-291, type 1 and secure with tape conforming to PPP-T-45, type III, grade A. Cushion each cable harness assembly by winding into a coil of proportionate dimensions and wrapping in paperboard conforming to PPP-P-291, type 1 and secure with tape conforming to PPP-T-45, type III, grade A. Place the wrapped items within a close-fitting fiberboard box conforming to PPP-B-636, W5c; fill all voids with paperboard conforming to PPP-P-291, type 1, to prevent movement. Place the technical literature packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level C. - Antenna AS-1304( )/ARN-30 shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited in transit storage.

5.2 Packing. - Packing shall be Level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents (See 6.2).

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5.2.1 Level A. -

5.2.1.1 Consolidation. - A quantity of Antennas AS-1304( )/ARN-30's, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box, conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load. - A quantity of containers, packed as specified in 5.2.1.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147, except that the pallet shall be softwood, conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.1.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type, PPP-B-621, style 4, class 2, or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more, and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto, except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B. -

5.2.2.1 Consolidation. - A quantity of Antenna AS-1304( )/ARN-30, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.

5.2.2.2 Palletized load. - A quantity of containers, packed as specified in 5.2.2.1, shall be palletized as specified in 5.2.1.2.

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5.2.2.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.2.1 shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, type IV, as specified in the appendix of the box specification. No further packing shall be required.

5.2.3 Level C. -

5.2.3.1 Consolidation. - A quantity of Antenna AS-1304( )/ARN-30, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the fiberboard boxes shall be class domestic.

5.2.3.2 Palletized load. - A quantity of containers, packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that the fiberboard cap shall be class domestic.

5.2.3.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.3.1 shall be used as the shipping container. No further packing shall be required.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use. - The antenna is intended to be used with Airborne Radio Sets to provide a capability for both fixed and rotary wing aircraft, in conjunction with ILS Localizer and VOR Navigational radio equipment. The nomenclature AS-1304( )/ARN-30 has been retained because it is the nomenclature for AN/ARN-30( ) Radio Set Antennas presently in use.

6.2 Ordering Data. - Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level of preservations, packaging and level of packing required for shipment. (Level A, Level B or Level C, See Section 5, paragraph 5.1 and 5.2)

c. The specified paragraphs of Section 5, which are applicable to the particular procurement, and when rough handling and functional tests are required.

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d. First Article Inspection. - Three sample units of each item cited in Section 3 are generally required so that lengthy environmental tests can be completed on two sample units while complete performance measurements can be made on the third sample unit. (See 3.2)

e. Marking and shipping of samples.

f. Place of final inspection.

g. Submission of the statement of treatment referenced in 3.10.4 should be after award of contract. This statement should be submitted to the contracting officer.

6.3 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AS-1304W/ARC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.4 Group C Inspection. - Approval to ship may be withheld, at the discretion of the Government, pending results of the corrective action specified in the contract or purchase order.

6.5 Definition of Packaging Levels. - Levels A & C packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during shipments and storage. (See 5.1)

6.6 Location of Operational Inspection. - It is desirable that the operational inspection in 4.11 be performed at a location that will minimize handling (which might cause damage to the units) after this inspection. It is recommended that the entire lot (including the sample units) be inspected immediately prior to packaging.

6.7 Verification Inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive Quality Assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product. (See 4.5)

Custodian
Army EL

Preparing Activity:
Army EL - 5826-A089

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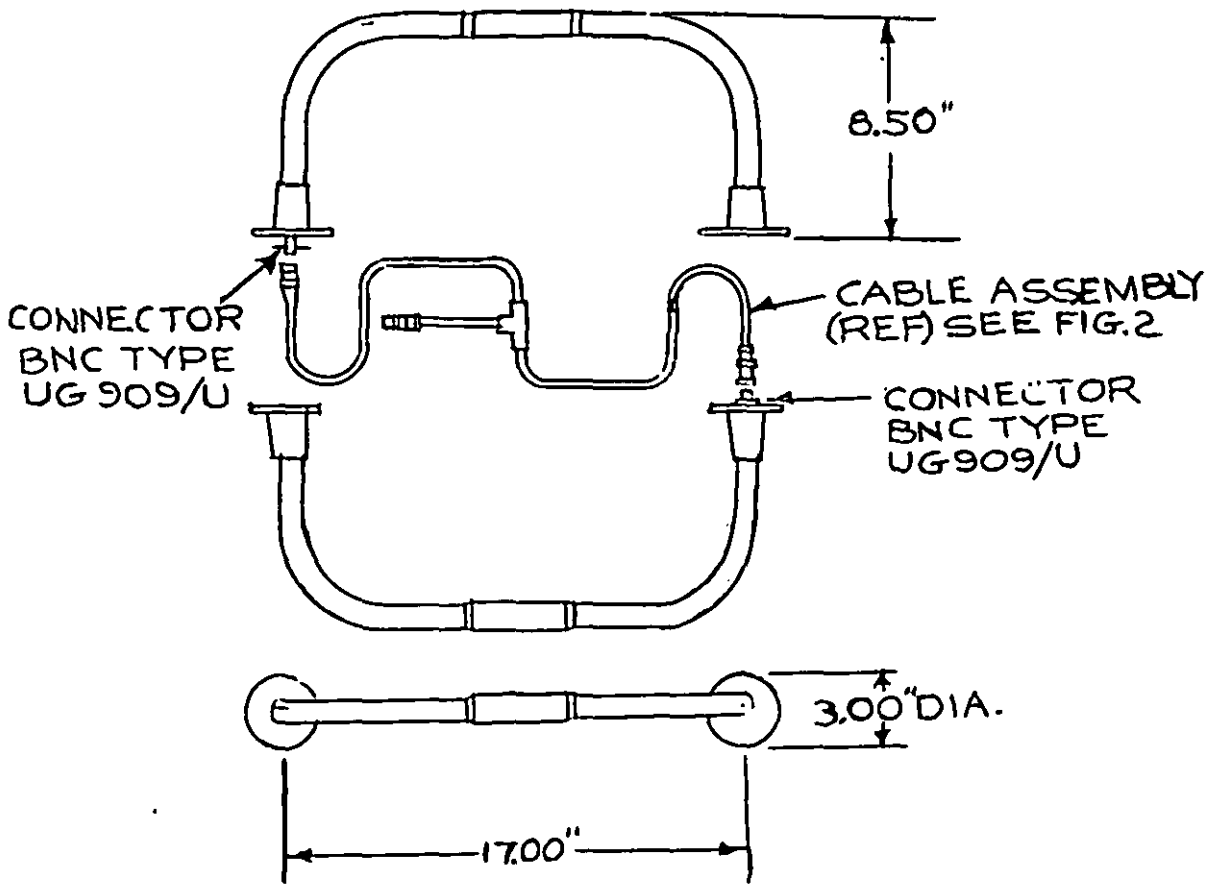


FIGURE 1. ANTENNA OUTLINE

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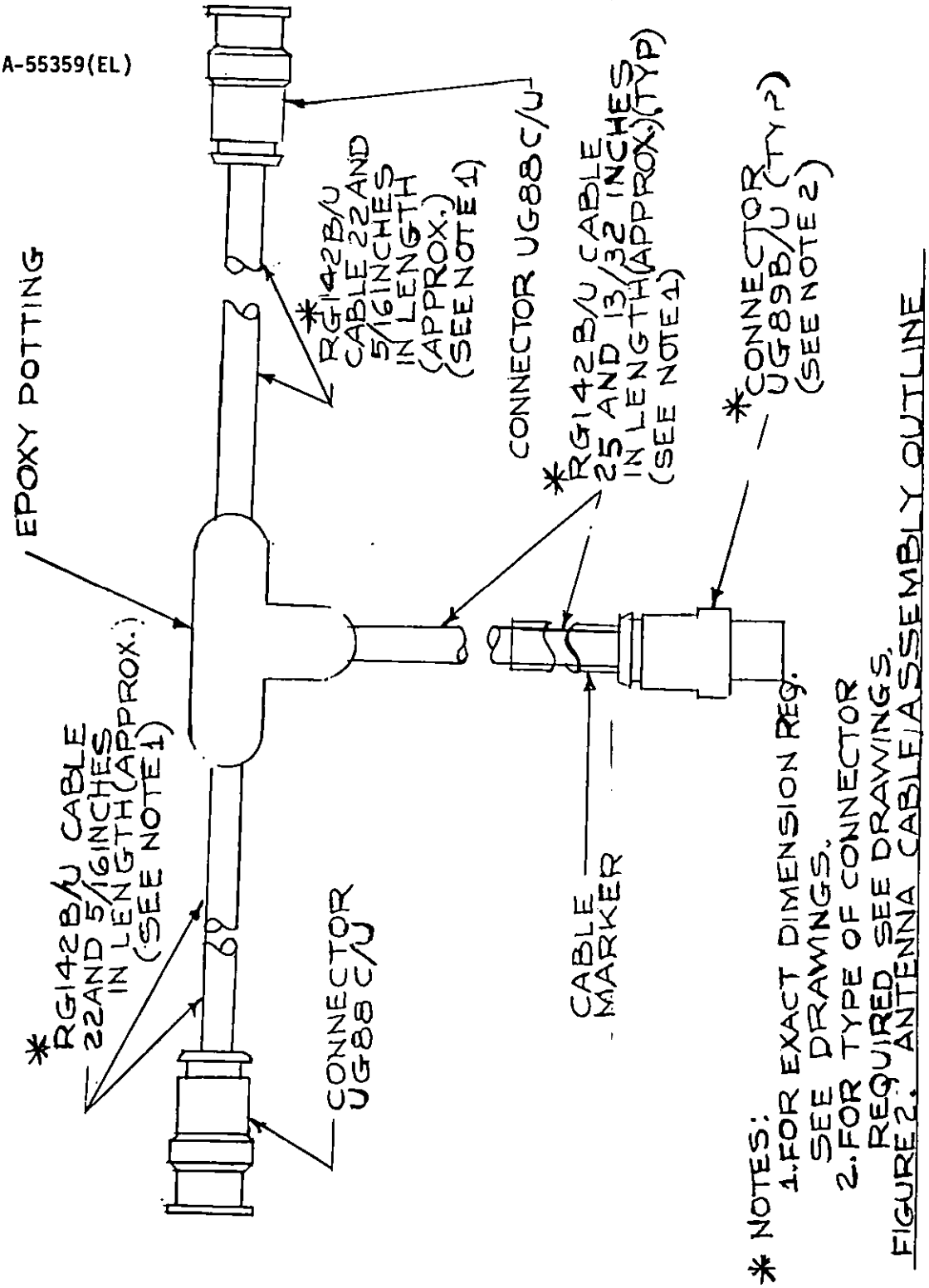


FIGURE 2. ANTENNA CABLE ASSEMBLY OUTLINE

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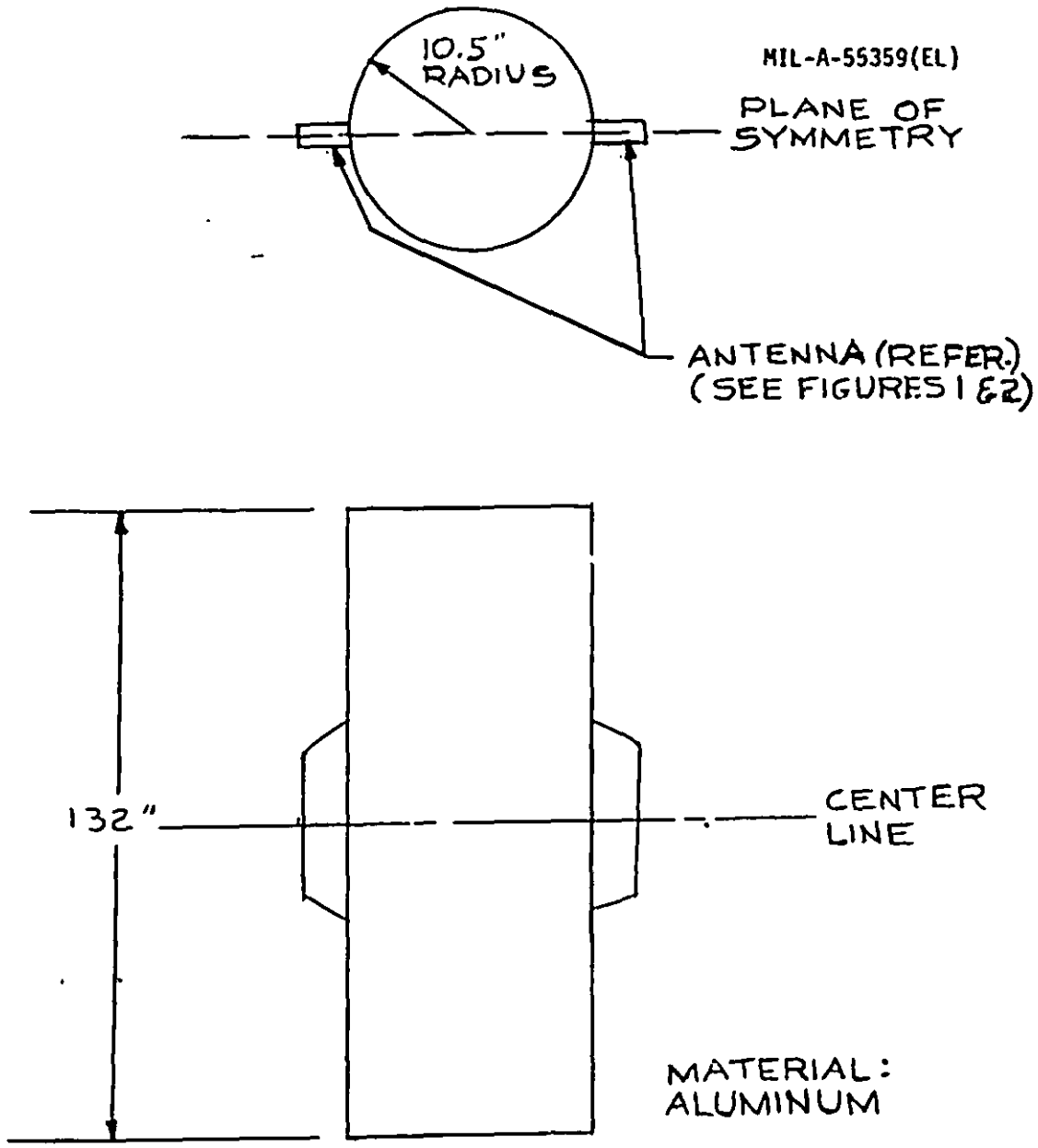


FIGURE 3. GROUND PLANE

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-A-55359(EL) ANTENNA AS-1304( )/ARN-30

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-PM 1068-68

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